PROBLEM STATEMENT #05 | Campus & Academic Operations

Academic Elective Bidding & Allocation System

1. Problem Context & Overview

University departments need an automated elective course selection mechanism where students allocate bidding credits across preferences and the engine solves allocation constraints while preventing timetable collisions.

Target Stakeholders / Actors: *Student, Academic Registrar*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]
Description: The system shall allow students to distribute 100 bidding credits across their ranked elective preferences and validate prerequisite course completions.
Sample Acceptance Criteria: Pass: Credits deduct accurately and prerequisites validated; Fail: Total allocated credits exceed 100 or unmet prerequisites bypassed.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]
Description: The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds.
Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**
  - >> Exactly 5 Functional Requirements (FR-001 to FR-005)
  - >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with
    ID, Type, Description, Priority, Acceptance Criteria, and Rationale.
- **UML Use-Case Diagram:**
  - >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.
- **Use-Case Flow Specification:**
  - >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
  - >> One Alternate Flow.